

CivicGuardian AI: An Agentic and Service-Oriented Framework for Autonomous Citizen Complaint Management

SOA Project Review-1

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Outline of Research Work

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Problem Statement

Existing citizen complaint management systems depend heavily on manual processing and human intervention, which makes it difficult to efficiently handle the increasing number of complaints related to public services and civic issues. Complaints may be incorrectly classified, sent to the wrong department, duplicated, delayed, or not given proper priority based on their urgency. Citizens often have limited ways to track their complaints and receive timely updates, while poor coordination between different civic departments can further increase resolution time. Existing systems also lack intelligent decision-making, automatic monitoring, and escalation of unresolved complaints, resulting in more manual work and lower efficiency. Therefore, there is a need for an intelligent, automated, and well-connected complaint management system that can improve the speed, accuracy, transparency, coordination, and overall quality of citizen complaint resolution.

Objectives

Develop an Agentic AI-based system capable of automatically managing, tracking, and resolving citizen complaints using a Service-Oriented Framework.

- Develop an intelligent system for citizen complaint management.
- Understand and classify complaints using AI and NLP.
- Identify the priority of each complaint.
- Send complaints to the correct department automatically.
- Detect duplicate or repeated complaints.
- Track the status and progress of complaints.
- Automatically escalate complaints when they are delayed.
- Provide regular complaint updates to citizens.
- Reduce manual work and improve complaint resolution time.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
S. P. Sanjay et al., 2026	Agentic AI-Driven Grievance Redressal Triage System for Multilingual Public Complaints	Agentic AI + LLM + RAG	Accurate complaint classification and routing	Tested on a limited complaint dataset
A. S. Najar et al., 2010	The Application of Service-Oriented Architecture in E-Complaint System	SOA + E-Government	Flexible integration of complaint services	Lacks modern AI automation
P. Aakash et al., 2026	CivicSense: AI-Based Citizen Complaint Analyzer	Generative AI + NLP	Automatic complaint analysis and prioritization	Further real-world deployment needed
S. Ganatra et al., 2025	Grahak-Nyay: Consumer Grievance Redressal through Large Language Models	LLM + RAG	Better understanding of citizen grievances	Mainly focused on consumer grievances

Research Gap Identification

Existing citizen complaint management systems have several limitations:

- Most systems depend heavily on manual complaint handling.
- Existing AI systems mainly focus on complaint classification.
- Limited use of Agentic AI for automatic decision-making.
- Poor coordination between different government departments.
- Limited automatic priority detection and complaint routing.
- Most systems do not automatically track and escalate delayed complaints.
- Few systems combine Agentic AI and Service-Oriented Architecture (SOA) in one framework.

Research Gap:

There is a need for an Agentic AI-based complaint management system that can automatically understand, prioritize, route, track, and escalate citizen complaints while connecting different civic services through a Service-Oriented Architecture.

Innovation, Creativity, and Novelty

Proposed Innovations:

- Integration of Agentic AI with Service-Oriented Architecture (SOA)
- Autonomous understanding and classification of citizen complaints
- Intelligent priority detection and smart department routing
- Automatic tracking and escalation of unresolved complaints
- Coordination between multiple AI agents and civic services
- End-to-end automated citizen complaint management

Novel Features:

- Intelligent complaint classification
- Automatic priority detection
- Smart department routing
- Duplicate complaint detection
- Real-time complaint tracking and status updates
- Automatic complaint escalation

Feasibility and Project Plan

